Mesh-to-Mesh Transfer Errors in Multistep Unsteady Flow Simulations

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Abstract

Mesh replacement changes both the spatial discretization and the solution histories used to advance an unsteady flow. This study examines how transfer accuracy, boundedness, integral preservation, and multistep-history consistency govern the disturbance introduced by remeshing and its subsequent aerodynamic response. Nearest-neighbor Taylor interpolation, barycentric interpolation, and continuous Galerkin projection are compared through smooth and discontinuous fields, repeated Euler remeshing, and stationary-airfoil calculations. Smooth tests recover the intended first- to third-order behavior. In repeated transfer, enforcing the global density integral to roundoff does not prevent substantial transition broadening, a result reproduced across three mesh-deformation realizations. Controlled restarts show that an error confined to the older history produces a finite next-step response even when the latest history is exact. Boundary enforcement also modifies transferred momentum. With two Reynolds-averaged Navier–Stokes closures, Galerkin projection reduces turbulence-integral defects by factors of approximately 37–247, but the smaller aerodynamic differences remain sensitive to time-step refinement and, for one closure, incomplete inner convergence. Transfer accuracy, admissibility, integral preservation, and multistep consistency therefore require separate assessment; none alone establishes the accuracy of the restarted calculation.

Keywords: Solution transfer, Remeshing, Conservative interpolation, Galerkin projection, Backward differentiation, Finite-volume methods

1. Introduction

Adaptive remeshing in an unsteady flow calculation creates a discrete restart each time the computational mesh is replaced. The numerical solution must be represented on the new mesh before time integration can continue, and repeated transfers can accumulate error over the simulation. Time-accurate mesh-adaptation frameworks therefore treat solution transfer as part of the accuracy problem itself, together with spatial and temporal discretization [1–3]. For a multistep scheme the issue is more restrictive than transferring the current state: every solution level required by the time integrator must be supplied consistently on the receiving mesh.

A transfer operator can satisfy several desirable properties, but these properties answer different questions. Interpolation order describes approximation of smooth fields; limiting controls the creation of new extrema; and conservative projection constrains selected integrals or moments. Common-refinement and Galerkin approaches provide accurate conservative transfer between nonmatching meshes [4, 5], while constrained formulations add further

discrete requirements such as bounds [6]. Conservative Galerkin transfer has also been coupled directly to time-accurate anisotropic adaptation, including recent high-order implicit formulations that introduce bounded transfer to improve robustness near discontinuities [3, 7]. None of these properties is interchangeable with another: preserving a domain integral does not determine the spatial distribution of the transferred field, and enforcing bounds can alter the approximation or projection that produced the raw target state.

The distinction is especially important after a multistep restart. Once the mesh changes, the receiving discretization advances from transferred histories rather than from a state generated by its own previous steps. An error confined to an older history can therefore perturb the next residual even when the latest history is exact. In addition, limiting, representation changes, positivity enforcement, and boundary conditions may modify the transferred field before it is actually used by the flow solver. Consequently, neither a latest-state interpolation norm nor a conservation defect alone determines the disturbance injected into the subsequent discrete evolution or into an aerodynamic output.

The question addressed here is therefore: *how do transfer accuracy, boundedness, integral preservation, and multistep-history consistency combine to determine the error introduced by mesh replacement and its subsequent aerodynamic response?* The study considers a vertex-centered finite-volume discretization advanced with the second-order backward differentiation formula (BDF2) on a fixed physical domain. Prescribed mesh replacements are used so that the transfer process can be examined independently of a solution-dependent adaptation criterion or moving-boundary effects. Three established transfer families—nearest-neighbor Taylor interpolation, barycentric interpolation, and continuous Galerkin projection—are compared, together with a bounded global correction that isolates the effect of enforcing a domain integral.

The numerical experiments are arranged to separate successive parts of this question. Smooth and discontinuous static fields establish approximation order, boundedness, and integral behavior without flow-solver effects. Repeated Euler remeshing then tests how those transfer properties accumulate. Independent Euler airfoil meshes permit the two BDF2 history levels to be perturbed separately, directly exposing their contribution to the restart response, while localized and repeated transfers examine the subsequent load evolution. Finally, fixed-body Reynolds-averaged Navier–Stokes (RANS) calculations add turbulence transport and viscous loads together with temporal, iterative, and mesh-sensitivity checks.

The evidence supports three conclusions that define the scope of the study. Enforcing a global integral can coexist with substantial cumulative spatial broadening; both BDF2 histories contribute directly to the restart transient, with boundary processing forming part of the effective transfer; and improved preservation of transferred integrals does not by itself establish a resolved aerodynamic advantage. The paper therefore evaluates transfer quality through complementary operator-level and solver-level diagnostics rather than seeking a universal ranking of interpolation methods.

2. Solution-transfer operators

The transfer methods are separated here into the construction of a raw target field and the constraints applied to the state ultimately returned to the flow solver. Three construction families are considered: nearest-neighbor Taylor interpolation (NN), barycentric

interpolation (BAR), and continuous Galerkin projection (GP). A numerical suffix denotes nominal smooth-field accuracy rather than polynomial degree; for example, BAR2 is the second-order barycentric variant, whereas GP2 uses piecewise-linear finite-element spaces. A bounded global correction (GC) is treated separately because it changes the integral of an existing target field rather than defining a new interpolation construction.

2.1. Representation and derivative reconstruction

Source and target meshes, A and B, cover the same discretized domain Ω. The nodal state on mesh M is U_M, and its finite-volume mass matrix is $M_M = \mathrm{diag}(w_i^M) \otimes I$, where w_i^M is the positive control-volume measure associated with vertex i, I is the identity on the state components, and $\otimes$ denotes the Kronecker product. For one scalar component ϕ, the solver integral is

$$Q_M(\phi_M) = \sum_i w_i^M \phi_{M,i}. \tag{1}$$

All methods return values at the flow-solver vertices. Mean-flow density, momentum, and total energy are transferred directly in conserved-variable form; E denotes specific total energy, so the energy inventory is ρE. Nodal interpolation is not control-volume averaging, even when Eq. (1) is used to assess its conservation.

Higher-order transfer requires derivatives recovered from neighboring values. A weighted least-squares fit uses

$$\phi_j - \phi_i \simeq g_i \cdot r_{ij} + \tfrac{1}{2} r_{ij}^T H_i r_{ij}, \qquad r_{ij} = x_j - x_i. \tag{2}$$

Here g_i and H_i approximate the gradient and Hessian of ϕ at source vertex i, and r_{ij} is the displacement to a neighboring vertex. A degree-one fit omits the Hessian. With smooth data, a full-rank scaled stencil, and controlled mesh geometry, the linear fit gives an $O(h)$ gradient error; the quadratic fit gives $O(h^2)$ gradient and $O(h)$ Hessian errors. Anisotropy, weighting, and stencil conditioning can degrade these estimates [8–11]. Third-order variants use the quadratic reconstruction, and their achieved order is verified numerically.

2.2. Nearest-neighbor and barycentric interpolation

Let $n(j)$ be the source vertex closest to target point x_j^B, and $r_j = x_j^B - x_{n(j)}^A$. Nearest-neighbor Taylor transfer gives

$$(\mathcal{I}_{AB}^{\mathrm{NN1}} \phi_A)_j = \phi_{A,n(j)}, \tag{3}$$

$$(\mathcal{I}_{AB}^{\mathrm{NN2}} \phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j, \tag{4}$$

$$(\mathcal{I}_{AB}^{\mathrm{NN3}} \phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j + \tfrac{1}{2} r_j^T H_{n(j)} r_j. \tag{5}$$

The variants respectively copy the donor value, include its local slope, and include its curvature. For $|r_j| = O(h)$ and sufficiently accurate derivatives, their smooth-field errors are $O(h)$, $O(h^2)$, and $O(h^3)$. The donor changes across source-node Voronoi boundaries, so the transferred value need not vary continuously with target position.

Barycentric interpolation instead uses the source simplex containing the target point. A conforming triangulation or tetrahedralization of the original mesh provides the search

geometry, without adding control-volume vertices. In spatial dimension d, a simplex has vertices $x^A_{i_r}$, $r = 1, \ldots, d+1$. Its barycentric coordinates satisfy

$$x^B_j = \sum_{r=1}^{d+1} \lambda_r x^A_{i_r}, \qquad \sum_r \lambda_r = 1, \qquad \lambda_r \geq 0. \tag{6}$$

The same geometric weights define the second-order interpolant and its gradient-corrected extension:

$$(\mathcal{I}^{\mathrm{BAR2}}_{AB}\phi_A)_j = \sum_r \lambda_r \phi_{A,i_r}, \tag{7}$$

$$(\mathcal{I}^{\mathrm{BAR3}}_{AB}\phi_A)_j = \sum_r \lambda_r \left[\phi_{A,i_r} + \tfrac{1}{2} g_{i_r} \cdot (x^B_j - x^A_{i_r})\right]. \tag{8}$$

BAR2 reproduces affine fields and is bounded by the containing-simplex values. For a quadratic field with exact vertex gradients, the half-gradient term cancels its interpolation remainder; BAR3 is consequently third-order consistent. Shared values and gradients give continuity across conforming simplex interfaces. Neither method imposes a source-to-target integral constraint.

Derivative limiting changes these properties independently of nominal order. Multiplying the Taylor corrections by $0 \leq \Phi_i \leq 1$ reduces NN2 or NN3 to NN1 when $\Phi_i = 0$, but reduces BAR3 to BAR2. The smooth verification is unlimited; discontinuous tests also use Barth–Jespersen (BJ) limiting [12]. A limiter derived from linear increments does not automatically bound a quadratic reconstruction at every evaluation point. Extrema must therefore be checked for the actual returned field.

2.3. Galerkin projection and cross-mesh integration

Continuous Galerkin projection determines a coupled target field rather than evaluating target vertices independently. Let ϕ^h_A be the piecewise-polynomial source representation and V^p_B the continuous degree-p Lagrange space on the target simplicial mesh. The projected field $\phi^h_B \in V^p_B$ satisfies

$$(\phi^h_B - \phi^h_A, v_B)_{L_2(\Omega)} = 0 \quad \text{for every } v_B \in V^p_B. \tag{9}$$

This is the best L_2 approximation of the source representation ϕ^h_A in V^p_B [4, 5]. Writing $\phi^h_B = \sum_j v_j N^B_j$, where N^B_j are target nodal basis functions and v_j their coefficients, gives

$$H_B v = b, \qquad (H_B)_{ij} = \int_\Omega N^B_i N^B_j \,\mathrm{d}x, \qquad b_i = \int_\Omega N^B_i \phi^h_A \,\mathrm{d}x. \tag{10}$$

The sparse positive-definite matrix H_B couples basis functions sharing target elements. It differs from the diagonal finite-volume matrix M_B used for time integration.

A target element can cross several source elements. Although a target basis function has one polynomial expression there, the source field changes expression at each source-element boundary. The cross-mesh load must therefore be integrated as

$$b_i = \sum_{K_B} \sum_{K_A : |K_A \cap K_B| > 0} \int_{K_A \cap K_B} N^B_i \phi^h_A \,\mathrm{d}x. \tag{11}$$

Here K_A and K_B are source and target simplices, and $|K_A \cap K_B|$ is their intersection measure. On each intersection, both expressions are fixed polynomials. These intersections form the common refinement and permit polynomial-exact quadrature: degree two for P^1/P^1 transfer and degree four for P^2/P^2 transfer on affine simplices [13, 14]. Complete, nonoverlapping coverage is required. The intersection geometry and coupled solve distinguish projection from the point searches used by NN and BAR.

GP2 uses P^1 source and target spaces. GP3 constructs a P^2 source representation, including internally reconstructed edge values, and solves for a complete quadratic target field. Only original-vertex values are then supplied to the flow solver; the edge construction and the consequence of this restriction are detailed in Appendix A. On mixed-element meshes the spaces are P^p on conforming simplicial subdivisions, not Q^p on the original elements. Nominal approximation orders are two and three under smoothness and mesh-stability assumptions; accuracy after vertex restriction is assessed separately.

2.4. Integral preservation, bounds, and the returned state

When the unconstrained target space contains constants, Eq. (9) preserves the integral of the complete finite-element field. It also preserves consistent coordinate moments represented by the test space. These identities do not imply conservation over arbitrary finite-volume cells, whose indicator functions generally lie outside a continuous finite-element space. Nor is the Galerkin mass-matrix inverse a local operator: unchanged geometry in a subregion does not guarantee unchanged projected values there.

For GP2, consistent and solver-lumped integrals coincide when $w_i^M = \int_\Omega N_i^M \, \mathrm{d}x$ on both meshes. For GP3, source reconstruction and target vertex restriction introduce additional representation differences. An exact quadratic projection can acquire a second-order lumped-integral error after vertex restriction (Appendix A). A third-order pointwise approximation therefore does not imply a third-order conservation defect in the solver representation.

The bounded global correction can enforce Eq. (1) for any base transfer. For a raw target field ϕ_B^r, the correction c satisfies

$$\sum_j w_j^B c_j = Q_A(\phi_A) - Q_B(\phi_B^r), \qquad \ell_j \le \phi_{B,j}^r + c_j \le u_j. \tag{12}$$

The nodal lower and upper bounds are ℓ_j and u_j. The defect is redistributed among nodes with available capacity, with saturated nodes fixed [15, 16]. The requested integral must lie within the weighted bounds. A small integral defect does not bound each correction, and bounds imposed separately on conserved components do not ensure positive pressure.

GP limiting uses source-vertex extrema in overlapping elements. The interval is widened by a dimensionless tolerance τ times the global source range, followed by clipping and redistribution preserving a selected lumped total. Hard limiting has $\tau = 0$; soft limiting permits a finite excursion. Once active, limiting need not preserve Galerkin orthogonality, moments, or best approximation. Boundary enforcement and positivity corrections can change the field further. Conservation and accuracy are consequently assessed at specified stages, rather than assigned to the final state from properties of the unconstrained projection.

3. Multistep response to a mesh change

The operator-level properties of a transfer become dynamically relevant through the solution histories supplied to the time integrator. The analysis therefore uses the native target-mesh trajectory as a common receiving reference, first identifying the exact BDF2 forcing generated by transferred histories and then relating that forcing to the solved state and aerodynamic outputs.

3.1. Native and transferred histories

Between mesh replacements the semi-discrete equations are

$$M_M \dot{U}_M + F_M(U_M, t) = 0. \tag{13}$$

Here t is physical time and F_M is the assembled spatial residual, including fluxes, source terms, and the imposed boundary treatment. The state includes every transported variable required by the time integrator, including turbulence variables for RANS. Boundary constraints must be imposed consistently on the histories and on the equations advanced in time. At a constant time step, the target BDF2 equation is

$$\Phi_B(Y; V, W) = \frac{M_B}{\Delta t}\left(\tfrac{3}{2}Y - 2V + \tfrac{1}{2}W\right) + F_B(Y, t^{n+1}) = 0. \tag{14}$$

Here V and W are the latest and older histories. The mesh is stationary during each time step. A continuously moving mesh would require compatible geometric histories and a discrete geometric conservation law [17, 18]; Eq. (14) is not intended as that general formulation.

Let U_A^k and U_B^k denote native trajectories for the same physical problem on two meshes. The transferred histories and their target-referenced discrepancies are

$$\widehat{U}_B^k = \mathcal{I}_{AB}(U_A^k), \qquad \varepsilon^k = \widehat{U}_B^k - U_B^k, \qquad k = n, n-1. \tag{15}$$

The effective map $\mathcal{I}_{AB}$ includes the processing required to form admissible histories. The discrepancy ε^k is not a pure interpolation error: it also contains the difference between the native source and target discretizations. Comparing with U_B^k provides a common receiving evolution operator, not an exact continuum reference. The nonlinear decomposition in Appendix A.1 makes this distinction explicit.

3.2. Exact history forcing and conditional state response

At a common trial state Y, changing only the two histories gives the exact identity

$$\begin{aligned} &\Phi_B(Y; U_B^n + \varepsilon^n, U_B^{n-1} + \varepsilon^{n-1}) - \Phi_B(Y; U_B^n, U_B^{n-1}) \\ &\qquad = -\frac{M_B}{\Delta t}\eta^n, \qquad \eta^n = 2\varepsilon^n - \tfrac{1}{2}\varepsilon^{n-1}. \end{aligned} \tag{16}$$

No linearity of the spatial residual or small-error assumption is required. Matching the trial state, boundary treatment, and residual sign convention is essential.

The solved-state difference $\delta^{n+1} = \widehat{U}_B^{n+1} - U_B^{n+1}$ obeys a different relation. Let r and $\widehat{r}$ be the algebraic residuals remaining after the native and perturbed physical steps. Subtraction and expansion about the native target state give

$$\begin{aligned} A_B\delta^{n+1} &= M_B\eta^n + \Delta t(\widehat{r} - r) - \Delta t\,\mathcal{N}_B(\delta^{n+1}) \\ A_B &= \tfrac{3}{2}M_B + \Delta t J_B, \qquad J_B = \frac{\partial F_B}{\partial U}. \end{aligned} \tag{17}$$

The Jacobian J_B is evaluated at the native target solution at t^{n+1}, and $\mathcal{N}_B$ is the nonlinear Taylor remainder. A locally Lipschitz Jacobian gives $\mathcal{N}_B(\delta) = O(\|\delta\|^2)$. If the step is locally invertible, the perturbation is small, and the algebraic errors are negligible, then

$$\delta^{n+1} = R_B\eta^n + O(\Delta t\|\eta^n\|^2), \qquad R_B = A_B^{-1}M_B. \tag{18}$$

The bound is local to the mesh, state, time step, and norm. Limiter switching and incomplete inner convergence can invalidate this smooth-response approximation without invalidating Eq. (16).

A superscript $\star$ denotes a computed steady reference state. Table 1 assigns the same spatial discrepancy $\varepsilon^\star = \mathcal{I}_{AB}U_A^\star - U_B^\star$ to the two histories independently. Writing the corresponding

Table 1: History perturbations used to separate the BDF2 contributions. The spatial discrepancy $\varepsilon^\star$ is the same within each method; the native case supplies the receiving-mesh reference.

Configuration	ε^n	ε^{n-1}	η^n
Native	0	0	0
Both histories	$\varepsilon^\star$	$\varepsilon^\star$	$3\varepsilon^\star/2$
Latest only	$\varepsilon^\star$	0	$2\varepsilon^\star$
Older only	0	$\varepsilon^\star$	$-\varepsilon^\star/2$

responses as δ_f, δ_n, and δ_{n-1}, local linearity predicts

$$\delta_n \simeq \tfrac{4}{3}\delta_f, \qquad \delta_{n-1} \simeq -\tfrac{1}{3}\delta_f, \qquad \delta_f \simeq \delta_n + \delta_{n-1}. \tag{19}$$

The older-only case tests whether a latest-state diagnostic misses a finite forcing. These collinear perturbations test directional response; they do not determine R_B for arbitrary history fields.

3.3. Aerodynamic response

For a differentiable target-mesh output G_B, define $\Delta G^{n+1} = G_B(\widehat{U}_B^{n+1}) - G_B(U_B^{n+1})$. With g_B^{n+1} the gradient of G_B with respect to the state at the native target solution, Eq. (18) yields

$$\Delta G^{n+1} = (g_B^{n+1})^T R_B\eta^n + O(\|\eta^n\|^2). \tag{20}$$

The load response depends on the spatial discrepancy weighted by the target dynamics and output sensitivity, not just its integral. In particular, preserving the source integral imposes, componentwise,

$$\mathbf{1}^T M_B\varepsilon^k = Q_A(U_A^k) - Q_B(U_B^k). \tag{21}$$

The right-hand side need not vanish because the native discrete states can have different integrals. Global correction therefore need not reduce the target-referenced state or load discrepancy.

After the first step, both the new solution difference and the remaining older history influence subsequent steps. A first-step load difference need not bound the later excursion. Disturbances generated away from the airfoil may produce a delayed response at an acoustic or convective scale, but neither the Galerkin mass-matrix inverse nor an implicit step has compact spatial support. The numerical experiments therefore use finite-amplitude arrival thresholds rather than the first nonzero signal.

4. Numerical experiments

The numerical experiments follow the causal chain developed above. Static fields first isolate approximation, boundedness, and integral behavior at the transfer stage. Repeated remeshing then tests accumulation, controlled airfoil histories isolate the multistep mechanism, and the RANS calculations determine whether transfer-level differences remain distinguishable once turbulence transport, viscous loads, and solver sensitivities are included.

4.1. Static transfer verification

Static transfer tests exclude flow-solver evolution so that operator properties can be measured directly. The smooth asymmetric field is

$$\phi = \rho_\infty \left[2 + 0.3e^{0.8x/L_x + 0.5y/L_y} \sin(2\pi x/L_x) \sin(2\pi y/L_y)\right]. \tag{22}$$

Here ρ_∞ is the reference density, and L_x and L_y are the coordinate length scales. Structured quadrilateral and independently generated mixed triangular/quadrilateral target meshes have nominal resolutions 64, 128, and 256; source resolution is half the target resolution. Relative nodal density errors are measured against the analytic field at target vertices. The effective spacing $h_{\text{eff}} = N_{\text{nodes}}^{-1/2}$ defines refinement ratios on the fixed domain. Pairwise orders are

$$p_{k,k+1} = \frac{\log(E_k/E_{k+1})}{\log(h_k/h_{k+1})}. \tag{23}$$

Here E_k is the relative nodal error and h_k the effective spacing at refinement level k. A least-squares fit of $\log E$ against $\log h$ summarizes each three-level sequence. The discontinuous field has density 1.2 outside and 3.6 inside a circular inclusion of radius $0.25 \min(L_x, L_y)$. The interface is not mesh-aligned. Unlimited and limited transfers are compared through overshoot, undershoot, and the source-to-target lumped-integral defect. Together, the smooth and discontinuous fields establish one-transfer properties before any error can accumulate through flow evolution.

4.2. Repeated Euler remeshing

The repeated-remeshing calculation asks whether the one-transfer properties measured above predict cumulative spatial fidelity. It uses a two-dimensional Euler calculation on $[0, 5] \times [0, 1]$. The uniform reference mesh has 501×101 vertices and spacing $h = 0.01$. The gas is ideal, with heat-capacity ratio $\gamma = 1.4$ and unit gas constant. Density, pressure, and

temperature are denoted by ρ, p, and T, and Cartesian velocity components by u and v. Initially,

$$(\rho, u, v, p) = \begin{cases} (1, 0, 0, 1), & x < 2, \\ (0.125, 0, 0, 0.1), & x \geq 2, \end{cases} \tag{24}$$

so the left and right temperatures are 1 and 0.8. The upper and lower boundaries are inviscid slip walls. At $x = 0$ and $x = 5$, characteristic boundary conditions prescribe total pressure and temperature $(p_0, T_0) = (1, 1)$ and $(0.1, 0.8)$, respectively, with direction $(1, 0)$ at both ends. The transient is compared with a fixed-mesh calculation using these same initial and boundary conditions, rather than with an exact Riemann solution.

The calculation advances 400 physical steps with $\Delta t = 10^{-3}$. Interior vertices receive fresh random displacements in x relative to the original mesh; boundaries remain fixed, and geometric distortion does not accumulate between successive meshes. For vertex i, let $a_{i,\min}$ be the smallest altitude of the incident corner triangles. The admissible streamwise displacement is bounded by $a_{i,\min}/s$, where the dimensionless deformation parameter s controls mesh perturbation amplitude. Values $s = 4$ and $s = 2$ are used; $s = 2$ permits twice the maximum displacement of $s = 4$.

Mesh replacement intervals of ten and two physical steps compare low- and high-frequency transfer. Both BDF2 histories are transferred at each mesh change. The low-frequency comparison contains 80 transfer evaluations; the high-frequency integral record contains 398. A fixed-mesh calculation supplies the reference. The comparison includes both transfer error and the change in spatial discretization; it is not an error against an exact Euler solution.

Density profiles are averaged in y with a common sampling procedure. The transition position is its 50% crossing, and its width is the distance between the 10% and 90% crossings. The differences Δx and Δw denote the transferred-calculation position and width minus their fixed-mesh reference values, expressed in cells of the $h = 0.01$ reference mesh. These metrics describe a resolved density transition rather than presume an independently identified sharp shock. Density and energy integral changes are evaluated immediately across each transfer, separately from physical evolution. With $I_\rho = Q_M(\rho_M)$, each signed increment ΔI_ρ is the target integral minus the source integral; $\sum \Delta I_\rho$ is their accumulated sum, not a sum of absolute errors. NN1, NN1+GC, BAR2, and GP2 are compared across three random deformation realizations at both amplitudes; other methods provide a broader frequency comparison at $s = 4$.

4.3. Euler airfoil and controlled BDF2 histories

The airfoil calculations test consequences that cannot be isolated in the repeated Riemann problem: the independent contribution of the two BDF2 histories and the evolution of a transfer disturbance into an aerodynamic output. Euler flow over the National Advisory Committee for Aeronautics (NACA) 0012 airfoil is calculated at freestream Mach number $M_\infty = 0.1$ and angle of attack $\alpha = 6°$, with $p_\infty = \rho_\infty = T_\infty = 1$, $a_\infty = \sqrt{1.4}$, $U_\infty = 0.118322$, and $\Delta t = 10^{-3}$. Here the subscript ∞ denotes freestream quantities, a_∞ is the sound speed, and U_∞ is the speed. The airfoil chord is denoted by c. The source and independent target meshes contain 16,255 and 22,508 vertices, respectively, with 32,077 and 44,583 triangles. Both retain the same 370 airfoil vertices and 63-vertex farfield polygon at $100c$, so the discretized surface is unchanged. Forces are pressure-only, with unit reference chord and area and a

moment reference at $x/c = 0.25$; lift and drag coefficients are denoted C_L and C_D. Nonzero steady drag is treated as a numerical load, not skin friction or shock-induced drag.

Each native steady solution is continued with equal initial BDF2 histories. The history combinations in Table 1 are then formed on the target mesh and compared with its native continuation at the same physical step. State differences use the nondimensional volume-weighted norm $\|v\|_{M_B} = (v^T M_B v)^{1/2}$ over the stored mean-flow components. Supplementary tests also report the unweighted Euclidean norm $\|v\|_2$; these norms weight the near-body and large farfield cells differently. Relative superposition defects are

$$e^U_{\mathrm{sup}} = \frac{\|\delta_f - \delta_n - \delta_{n-1}\|_{M_B}}{\|\delta_f\|_{M_B}}, \qquad e^{C_L}_{\mathrm{sup}} = \frac{|\Delta C_{L,f} - \Delta C_{L,n} - \Delta C_{L,n-1}|}{|\Delta C_{L,f}|}. \tag{25}$$

State-norm ratios are nonnegative; force ratios retain their sign.

The solver reports $-\Phi_B$. Its initial residuals are compared before an update using two common-trial pairs: older-only minus native at $U_B^\star$, and latest-only minus both-histories at $\widehat{U}_B^\star$. Their differences in η are $-\varepsilon^\star/2$ and $+\varepsilon^\star/2$, respectively. Double-precision residuals and a wall-normal/tangential split separate temporal forcing from slip-wall enforcement. Additional cancellation and amplitude tests are described in Appendix C.

A repeated-remeshing experiment perturbs vertices at wall distances between $0.01c$ and $20c$, while retaining the boundaries. NN2 and GP2 use the same two deformation realizations, with 39 mesh replacements followed by evolution in a 400-step interval. Load differences are formed against the corresponding fixed-mesh trajectory. Slopes per 100 physical steps are descriptive finite-window measures, not estimates of a long-time drift or independent-sample uncertainty.

A separate single-transfer experiment perturbs narrow bands beginning at wall distances $0.05c$, $0.10c$, and $0.20c$. Fixed pressure shells measure the root-mean-square (RMS) pressure difference, and force onset is defined by a fraction of the response maximum. For a shell at r inward of a band at d, the low-Mach acoustic scale is $(d-r)/(a_\infty \Delta t) \simeq 845(d-r)$ steps. Immediate disturbances extend up to approximately $0.08c$ inward of the displaced band; those shells are excluded from the propagation fit. Results are interpreted as threshold-dependent propagation, with a common fixed-mesh reference, rather than strict locality of interpolation.

4.4. Fixed-body RANS and numerical sensitivity assessment

The RANS study asks whether the transfer-level distinctions identified in the inviscid experiments remain visible when turbulence variables, viscous forces, and imperfect numerical resolution are introduced. The RANS case retains $M_\infty = 0.1$ and $\alpha = 6°$, with chord Reynolds number $Re_c = 6 \times 10^6$ and the Spalart–Allmaras (SA) and shear-stress-transport (SST) closures [19, 20]. The coarse mesh has 23,271 vertices, 9,686 quadrilaterals in the wall layer and wake region, 26,851 triangles, and 256 airfoil vertices. The medium and fine meshes contain 32,102 and 48,228 vertices, respectively; all three meshes retain the same first off-wall spacing. The farfield is at $100c$. The first off-wall spacing is approximately $5 \times 10^{-6}c$, and the quadrilateral region extends to about $0.11c$. This is a remeshing experiment about a nominally steady viscous state, not a turbulence-model validation.

The calculations use the Stanford University Unstructured (SU2) solver [21]. Mean-flow convection uses the Harten–Lax–van Leer contact (HLLC) flux with a monotonic upstream-centered scheme for conservation laws (MUSCL) reconstruction; turbulence transport uses

scalar upwinding without MUSCL reconstruction. BDF2 physical steps are solved by dual-time iteration at a Courant–Friedrichs–Lewy (CFL) number of 6. Linear systems are solved with flexible generalized minimal residual (FGMRES) iterations and incomplete lower–upper (ILU) preconditioning. The nominal inner-iteration limit is 800 iterations. After ten iterations, the stopping test checks $\log_{10}\mathrm{rms}(r_\rho) \le -12$, where r_ρ is the nodal density-equation residual in the solver normalization. The dimensional physical step $\Delta t_0 = 10^{-3}$ s corresponds to the nondimensional value $\Delta t_0^* = 0.347224$. A superscript $*$ denotes quantities normalized with the solver reference scales. With $U_\infty^* = 0.1$, the 400-step interval spans approximately 13.9 chord flow-through times, where convective time is measured by tU_∞/c. A first-order startup step is ten times smaller. Each closure has a no-remeshing reference continued for 800 steps from matching histories.

On the coarse mesh, the steady SST solution reaches the density-residual target, whereas the SA solution reaches its 20,000-iteration limit at -11.2, with lift and drag stationary to 10^{-7} over the final 2,000 iterations. The steady SST solution on the medium mesh stops at approximately -11.1. These residual levels and the reference continuations are distinguished from convergence of the remeshed physical steps.

The same prescribed mesh sequences are used for NN2 and limited GP2. At each replacement, 12,602 vertices may move and 10,669 are fixed. No vertex with solver wall distance below $0.01c$ moves; the geometric mask uses distance to the nearest airfoil vertex, and part of the outer quadrilateral layer can move. The nominal RANS deformation uses $s = 4$, and a minimum corner-area ratio of 0.2 preserves valid cells. The boundaries remain fixed. Mesh replacement occurs every ten physical steps, independently of the evolving solution.

Mean-flow and turbulence histories are transferred at both required time levels. For SA, $\widetilde{\nu}$ is the transported working variable. For SST, k and ω denote turbulent kinetic energy and specific dissipation rate; their density-weighted values are transferred before primitive-variable positivity floors are applied. Wall distance and eddy viscosity are recomputed. Turbulence integrals are not invariants of their subsequent transport equations. The SST transfer-integral diagnostic is evaluated before the final positivity floor and does not, by itself, establish conservation of the fully processed restart.

The pressure and viscous contributions to total drag are denoted by $C_{D,p}$ and $C_{D,v}$, respectively. For an aerodynamic coefficient C_Q and transfer method m, define the remeshing-induced response and paired method difference by

$$D_{m,Q}(t) = C_{Q,m}(t) - C_{Q,\mathrm{ref}}(t), \qquad P_Q(t) = D_{\mathrm{NN2},Q}(t) - D_{\mathrm{GP2},Q}(t). \tag{26}$$

Pressure and viscous contributions use identical surface quadrature for the remeshed and reference calculations. Over N common physical sample times t_i, the RMS of a scalar history q is $\mathcal{R}(q) = (N^{-1}\sum_i q(t_i)^2)^{1/2}$. For a modified numerical setting c, relative to the nominal setting denoted by subscript 0, sensitivity is measured by

$$S(q;c) = \frac{\mathcal{R}(q_c - q_0)}{\mathcal{R}(q_0)}. \tag{27}$$

This measures the difference between histories, not the fractional change of their RMS amplitudes. The latter is $[\mathcal{R}(q_c) - \mathcal{R}(q_0)]/\mathcal{R}(q_0)$.

The sensitivity assessment uses a common 38-sample window containing three mesh replacements from one deformation realization. Inner-iteration limits are increased to 1600

for both closures and to 6400 for SST. Time-step refinement uses $\Delta t_0/2$ for both closures and $\Delta t_0/4$ for SA, with identical displacement fields applied at matching physical times and a separate no-remeshing reference for each step size. The medium and fine steady meshes assess sensitivity of the absolute loads. A short SA remeshing comparison on the medium mesh uses the same displacement statistics and amplitude limit but an independent nodal deformation field; it is treated as a cross-mesh robustness test rather than an isolated spatial-convergence study.

Computational assistance. Generative artificial intelligence (AI) tools, ChatGPT (OpenAI) and Claude (Anthropic), assisted the design of verification tests, code development, and numerical post-processing. The tabulated values and data-driven figures are obtained from numerical outputs through the definitions given above, not from generative model predictions. *[Authors: specify the model versions and verify the scope of research assistance and the independent checks performed.]*

5. Results

The results are interpreted in the same order as the numerical mechanisms are introduced: first the intrinsic transfer properties, then their accumulation, then the multistep restart response, and finally the aerodynamic consequences under Euler and RANS evolution.

5.1. Interpolation order verification

Before flow-response differences are interpreted, the transfer operators must recover their intended smooth-field accuracy. Figure 1 shows the corresponding refinement results. NN1 is first order; NN2, BAR2, and GP2 are consistent with second order; NN3 is consistent with third order. BAR3 and GP3 have fitted slopes above three over the three-level interval, approaching approximately 3.2 on the finest pair. This is finite-range convergence behavior, not evidence of a higher designed order. On the finest mixed grid, the relative errors are approximately 2.6×10^{-7} for BAR3, 3.4×10^{-7} for GP3, and 2.6×10^{-6} for NN3. These differences resolve smooth interpolation performance on the tested mesh family; they do not predict a corresponding ranking of restarted flow solutions.

5.2. Boundedness and conservation

Smooth-field order does not determine behavior near a discontinuity. The circular-inclusion test therefore separates boundedness from integral preservation. Table 2 compares representative transfers at the middle structured resolution, $64 \rightarrow 128$. Unlimited NN2 exceeds the density plateaus by 0.8031 above and 0.4157 below. BAR3 reduces these excursions but remains unbounded. Barth–Jespersen limiting removes the measured new extrema; where it suppresses the NN2 reconstruction, however, the integral defect increases toward the NN1 value. Global correction then restores the integral without reintroducing an extremum. Hard-limited GP2 is bounded and conservative in this representation, whereas vertex-only GP3 retains a nonzero lumped-integral defect. The methods therefore cannot be characterized by one combined measure of order, boundedness, and conservation.

For a separate smooth transfer, global correction reduces the density-integral defect to order 10^{-15} while changing the density L_1 error by at most 0.8%. Its one-step accuracy penalty is small in that test. Whether the same scalar constraint controls repeated spatial distortion is examined next.

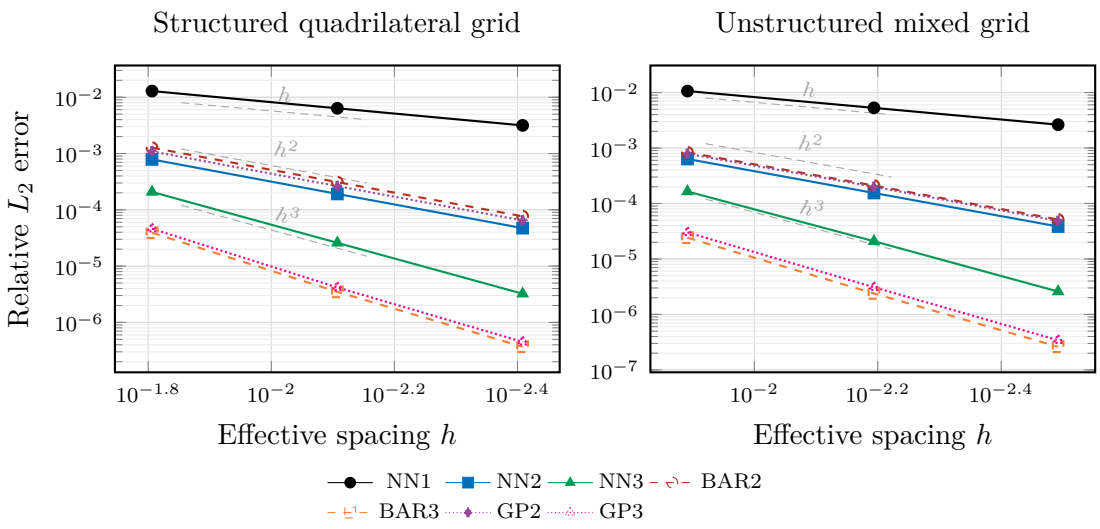


Figure 1: Smooth-field transfer accuracy on structured quadrilateral (left) and unstructured mixed (right) target meshes. The ordinate is the relative nodal density L_2 error; h is the count-based effective spacing. NN, BAR, and GP denote nearest-neighbor Taylor interpolation, barycentric interpolation, and Galerkin projection, with numerical suffixes indicating nominal accuracy order. Gray lines are first-, second-, and third-order slope guides, not fitted curves.

Table 2: Representative discontinuous-field transfers. Overshoot and undershoot are excursions beyond the exact density range $[1.2, 3.6]$; the last column is the source-to-target defect in the solver's lumped density integral. BJ denotes Barth–Jespersen limiting and GC the bounded global conservation correction.

Transfer	Overshoot	Undershoot	Integral defect
NN2, unlimited	0.8031	0.4157	-5.26×10^{-5}
NN2, BJ	0	0	-4.46×10^{-3}
NN2, BJ+GC	0	0	2.95×10^{-14}
BAR3, unlimited	0.1766	0.1811	-8.80×10^{-7}
BAR3, BJ+GC	0	0	5.59×10^{-14}
GP2, hard limited	0	0	3.44×10^{-15}
GP3, hard limited, vertex output	0	0	-4.83×10^{-7}

5.3. Accumulated broadening under repeated Euler remeshing

The static tests establish one-transfer properties but do not show whether those properties control accumulated remeshing error. The repeated Euler calculation addresses that question. With mesh replacement every ten physical steps, every method locates the density transition within 0.14 reference-grid cell of the fixed-mesh reference. Width changes are also small: 0.57 cell for NN1, 0.25 for BAR2, and at most 0.01 in magnitude for the remaining variants. Increasing the replacement frequency changes the width much more than the position (Table 3). At $s = 4$, NN1 broadening grows to 21.41 cells and BAR2 broadening to 3.24 cells, while GP2 has no width change to the reported precision. At $s = 2$, NN1 and BAR2 broaden by 26.36 and 22.91 cells, respectively, compared with 0.06 for GP2.

The hierarchy repeats across three random realizations. At $s = 4$, excess widths range from 21.41 to 22.89 cells for NN1, 3.12 to 3.24 for BAR2, and 0 to 0.01 for GP2. At $s = 2$ the

Table 3: Broadening of the density transition relative to the fixed-mesh reference. Width increases and the largest absolute position error among the listed methods are expressed in cells of the $h = 0.01$ reference mesh. GC is the global correction; $s = 2$ permits twice the displacement amplitude of $s = 4$.

Condition	NN1 width	NN1+GC width	BAR2 width	GP2 width	max. $\lvert\Delta x\rvert/h$
80 transfers, $s = 4$	0.57	–	0.25	0.00	0.14
398 transfers, $s = 4$	21.41	18.36	3.24	0.00	0.15
398 transfers, $s = 2$	26.36	25.65	22.91	0.06	0.91

respective ranges are 24.98–27.14, 22.90–22.98, and 0.03–0.06 cells. Position errors remain below 0.5 cell for these uncorrected comparisons. The dominant degradation is thus broadening rather than systematic translation of the feature.

Global correction isolates the distinction between integral and spatial error. Across all corrected NN1 realizations, the absolute cumulative density-integral defect does not exceed 9.4×10^{-14}, yet excess width remains 18.36–21.28 cells at $s = 4$ and 23.80–26.32 cells at $s = 2$. Correcting the total does not recover the lost profile. The result does not establish a unique benefit of Galerkin projection: NN2, NN3, BAR3, and GP3 also have much smaller width changes than NN1 in the $s = 4$ comparison. Appendix B retains those alternatives and the complete realization data.

5.4. Response to the two BDF2 histories

The repeated-remeshing results establish accumulation but do not identify how a multistep integrator responds to errors in its stored histories. A target-referenced state norm alone cannot isolate that mechanism: on the independent Euler airfoil meshes, second- and third-order transfers cluster near 7.0×10^{-4}, with less than 3% spread, whereas NN1 gives 1.47×10^{-3}. These values include the difference between the native discrete equilibria and therefore cannot be read as pure interpolation errors or used to rank the higher-order transfers.

The controlled-history experiment removes that ambiguity by perturbing the two BDF2 levels separately. Figure 2 shows a finite older-only lift response for every transfer, although the latest history is exact. For GP2, the older-only lift difference is $+2.539 \times 10^{-3}$, compared with -7.575×10^{-3} when both histories are perturbed. The latest-state discrepancy alone therefore misses a nonzero forcing of the next physical step.

Across all seven transfer variants, the latest/full state-norm ratio is 1.3330–1.3333 and the older/full magnitude is 0.3335–0.3341, close to the BDF2 weights in Eq. (19). Volume-weighted state superposition defects are 1.8–3.5%, and lift superposition defects are 0.28–1.49%. These measurements support the predicted directional response, not an independently validated linear solution operator for arbitrary perturbations.

The measured response ratios establish the temporal weighting, but they do not yet test whether the transferred history enters the discrete equations exactly as predicted. The common-trial residual comparison provides that check and identifies an additional boundary effect. NN and BAR satisfy the history-forcing identity to approximately 10^{-7} relative error. For GP2 and GP3, the larger whole-vector mismatch is confined to wall-normal momentum at the 370 slip-wall vertices. The normal mismatch equals the corresponding transferred forcing, while tangential differences are of order 10^{-11}. At the projected boundary rows, the same boundary operation must be applied to the predicted history forcing; the interior and

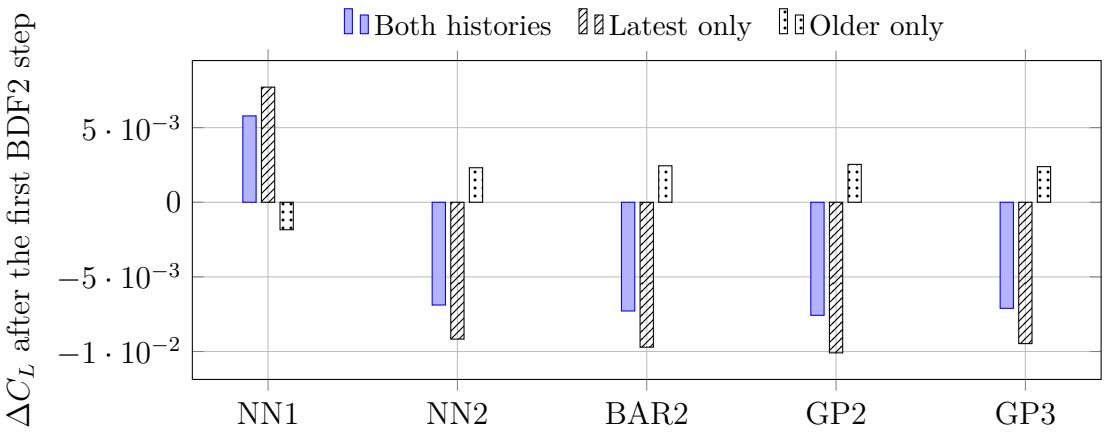


Figure 2: First-step lift-coefficient difference ΔC_L after transfer to the target Euler airfoil mesh, relative to its native continuation. NN, BAR, and GP denote nearest-neighbor Taylor interpolation, barycentric interpolation, and Galerkin projection; suffixes indicate nominal accuracy order. Within each method, the same spatial discrepancy is assigned to both histories or to one history only. The opposite signs of the latest-only and older-only responses are consistent with the second-order backward differentiation weights.

tangential components then recover the identity. The distinction is between the unconstrained projected history and the history admitted by the receiving equations, not a failure of the BDF2 identity.

Two supplementary tests delimit how far the local-response interpretation can be taken. Cancellation tests produce very small first-step force differences, but not comparably small volume-weighted state differences. Reducing the imposed amplitude decreases Euclidean-state and lift superposition remainders, whereas the relative volume-weighted remainder increases. Both diagnostics are retained in Appendix C; neither an iterative floor nor a uniform quadratic remainder is inferred from these observations.

Finally, the same controlled restart shows that exact preservation of a global integral is not a proxy for the load response. Global correction changes the NN1 full-history lift response from 5.7873×10^{-3} to 5.7924×10^{-3} and the BAR2 response from -7.2852×10^{-3} to -7.2850×10^{-3}. In these cases the integral constraint makes little difference to the load response, consistent with the spatial weighting in Eq. (20).

5.5. Accumulation and delayed Euler load response

The controlled-history experiment establishes an immediate restart response, but a first-step load difference need not describe its subsequent evolution. The repeated Euler airfoil calculation therefore measures accumulation over a finite window and shows a method-dependent drag trend. The GP2 slopes are 6.84×10^{-6} and 6.38×10^{-6} per 100 physical steps for the two realizations; NN2 gives 3.95×10^{-5} and 3.84×10^{-5}. The ratio of the two-realization means is approximately 5.9. This is a repeatable contrast for the tested discretization and interval, not a universal accuracy or cost ranking. The 400-step Euler interval is only about 0.047 chord flow-through times, and lift trends are less consistent between realizations.

A separate localized-transfer experiment tests the complementary question of delayed propagation from the perturbed region to the airfoil. On the longer $0.20c$-band case, pressure-shell propagation speeds inferred from 10–75% amplitude thresholds range from 1.09 to

1.27, bracketing $a_\infty = 1.183$. This range measures detector sensitivity for one event, not an ensemble confidence interval. It supports an acoustic-scale delayed response outside the immediate transfer footprint. Separately, the NN1 lift difference peaks three steps after the first post-transfer sample, at 4.66 times its initial value; the sampled second- and third-order responses instead peak at the first step and relax thereafter.

5.6. RANS transfer integrals and aerodynamic resolution

The RANS comparison asks whether a clear transfer-level difference survives as a resolved aerodynamic difference once turbulence transport, viscosity, and numerical sensitivities are present. At the transfer stage, GP2 preserves the monitored integrals more closely than NN2 (Table 4). Maximum density and total-energy defects differ by roughly two orders of magnitude. For the turbulence inventories the NN2/GP2 ratios are approximately 247 for SA and 37 for both SST quantities. These are instantaneous transfer defects at the measurement stage described in Sec. 4.4, before the final SST positivity floor; they are not conservation errors accumulated through the turbulence equations.

Table 4: Largest absolute source-relative integral change across the repeated RANS history transfers. Each column is maximized separately over the methods' two realizations. The SST values precede the final positivity floor. The inventory column identifies the integrand; the last column is the ratio of the separate maxima.

Closure	Inventory	NN2	GP2	NN2/GP2
SA	ρ	$8.41 \cdot 10^{-11}$	$4.57 \cdot 10^{-13}$	184
SA	ρu	$3.78 \cdot 10^{-9}$	$5.76 \cdot 10^{-11}$	66
SA	ρv	$1.65 \cdot 10^{-8}$	$8.46 \cdot 10^{-11}$	195
SA	ρE	$9.54 \cdot 10^{-11}$	$5.83 \cdot 10^{-13}$	164
SA	$\rho\tilde{\nu}$	$1.7 \cdot 10^{-6}$	$6.88 \cdot 10^{-9}$	247
SST	ρ	$8.25 \cdot 10^{-11}$	$4.68 \cdot 10^{-13}$	177
SST	ρu	$3.65 \cdot 10^{-9}$	$2.89 \cdot 10^{-11}$	126
SST	ρv	$1.64 \cdot 10^{-8}$	$7.78 \cdot 10^{-11}$	211
SST	ρE	$9.37 \cdot 10^{-11}$	$6.75 \cdot 10^{-13}$	139
SST	ρk	$2.22 \cdot 10^{-6}$	$6 \cdot 10^{-8}$	37
SST	$\rho\omega$	$7.98 \cdot 10^{-8}$	$2.15 \cdot 10^{-9}$	37

That strong integral separation does not carry directly into the loads: over the repeated-remeshing interval, total-drag RMS deviations from the no-remeshing references range from 6.07×10^{-5} to 6.79×10^{-5}. Paired NN2–GP2 differences have RMS 0.77×10^{-5} to 1.11×10^{-5}, and the ordering of the methods changes with closure and realization. Similar aggregate deviations do not imply identical histories or aerodynamic equivalence.

Because the paired aerodynamic difference is much smaller than the individual remeshing responses, its interpretation depends on numerical resolution. The time-step sensitivity tests distinguish the overall disturbance from that smaller paired difference (Table 5). Halving the step changes the time-aligned individual lift histories by 15.1–16.5% of their nominal-setting RMS for SA and 19.9–20.5% for SST; drag changes are 8.0–10.2% and 10.3–10.8%, respectively. Their RMS amplitudes change much less. In contrast, the paired lift and drag differences change by 75.2% and 128.7% for SA, and by 146.4% and 110.3% for SST. The SA quarter-step comparison increases the change of the paired difference further. Thus the overall disturbance

remains of comparable magnitude, but neither the detailed transient nor a method ranking is established as time-step converged.

Table 5: Time-step sensitivity at common physical times over the three-replacement RANS sensitivity window. Values are $100S$ from Eq. (27). Individual columns give the larger NN2 or GP2 sensitivity; paired columns refer to their difference. These are RMS differences between histories, not percentage changes in their RMS amplitudes.

		Individual histories (%)		Paired difference (%)	
Closure	Step	C_L	C_D	C_L	C_D
SA	$\Delta t_0/2$	16.5	10.2	75.2	128.7
SA	$\Delta t_0/4$	18.8	15.0	131.5	171.0
SST	$\Delta t_0/2$	20.5	10.8	146.4	110.3

Iterative convergence provides a second resolution check. SA reaches the density-residual target at every reduced-step sample. Increasing the inner-iteration limit at the nominal physical step has a smaller, but nonzero, effect on the paired difference. SST is less conclusive: most moved-grid samples in the increased-iteration calculations still reach the iteration limit, and the $\Delta t_0/2$ calculation is also incompletely converged. Its observed time-step sensitivity therefore includes possible iterative-error contributions. Appendix D quantifies these limits, including the more sensitive friction contribution.

Spatial sensitivity must be interpreted separately from the common-mesh method comparison. The coarse, medium, and fine steady viscous meshes show sensitivity of the absolute loads, but do not establish an asymptotic refinement sequence. That comparison is not an error bound on the paired NN2–GP2 difference on a common mesh. Likewise, the medium-mesh remeshing comparison changes the realized nodal disturbance and cannot isolate mesh convergence of the transient. Within these limits, RANS extends the integral comparison to turbulence variables and viscous flow, while providing no resolved aerodynamic advantage for either transfer method.

6. Conclusions

The results show that the quality of a mesh-to-mesh restart cannot be reduced to a single transfer metric. Approximation order, boundedness, integral preservation, admissibility of the returned state, and consistency of all multistep histories constrain different parts of the restart error, and their effects must be separated before an aerodynamic consequence can be assigned to a transfer method.

The repeated Euler remeshing provides the clearest counterexample to judging transfer quality from a global conservation measure alone. Correcting the density integral to roundoff does not prevent substantial transition broadening across three deformation realizations. Conversely, several higher-order pointwise and projection methods retain the profile much more closely in the milder deformation case. The relevant distinction is therefore between preservation of selected global quantities and preservation of the spatial state needed by the subsequent discretization.

The controlled BDF2 restarts identify the temporal mechanism. An error confined to the older history produces a finite next-step response even when the latest history is exact,

and the measured directional response follows the BDF2 history weights within the observed superposition errors. The residual tests further show that boundary enforcement changes the effective history presented to the receiving equations. A small latest-state error, a small first-step load difference, or an exactly corrected global integral is therefore insufficient on its own to characterize restart fidelity.

The aerodynamic calculations then delimit how far operator-level improvements can be translated into output accuracy. GP2 produces a smaller finite-window Euler drag trend than NN2 and much smaller RANS transfer-integral defects, but the RANS load difference between the two methods is not resolved with respect to the available temporal and iterative sensitivity tests; SST is additionally affected by incomplete inner convergence. The evidence therefore supports a framework for assessing transfer-induced error rather than a universal ranking of transfer schemes: the transferred spatial state, all required time histories, boundary admissibility, and the numerical resolution of the receiving calculation must be evaluated together.

Appendix A. Representation and reference-state identities

GP3 reconstructs the source value at the midpoint of edge (i, j) as

$$\phi_{ij} = \tfrac{1}{2}(\phi_i + \phi_j) - \tfrac{1}{8} d_{ij}^T \overline{H}_{ij} d_{ij}, \qquad d_{ij} = x_i - x_j, \qquad \overline{H}_{ij} = \tfrac{1}{2}(H_i + H_j). \tag{A.1}$$

For a quadratic field, the curvature term is exactly the difference between the endpoint average and midpoint value. Quadratic projection can consequently be exact even though restriction changes its solver integral. For example, $f(x) = x^2$ on a uniform partition of $[0, 1]$ belongs to the quadratic finite-element space, but integration of only its vertex values with linear nodal weights gives

$$Q_h(f) = \frac{1}{3} + \frac{h^2}{6}. \tag{A.2}$$

The second-order integral discrepancy arises solely from the representation change, not an error in the complete projection.

Appendix A.1. Discrepancy relative to a native target state

Let $\Pi_M u^k$ represent a common exact field in the solver representation, with native error $e_M^k = U_M^k - \Pi_M u^k$. For the complete, possibly nonlinear, transfer map,

$$\begin{aligned} \varepsilon^k = {} & \underbrace{\mathcal{I}_{AB}(\Pi_A u^k) - \Pi_B u^k}_{\text{transfer consistency}} \\ & + \underbrace{\mathcal{I}_{AB}(U_A^k) - \mathcal{I}_{AB}(\Pi_A u^k) - e_B^k}_{\text{source and target discretization contributions}} . \end{aligned} \tag{A.3}$$

Only a linear map permits the second line to be written $\mathcal{I}_{AB} e_A^k - e_B^k$. A load difference likewise separates as

$$\begin{aligned} G_B(\widehat{U}_B^k) - G_A(U_A^k) = {} & G_B(U_B^k) - G_A(U_A^k) \\ & + G_B(\widehat{U}_B^k) - G_B(U_B^k). \end{aligned} \tag{A.4}$$

The first difference is native mesh sensitivity; the second is the target-referenced restart response. Neither alone is an independently known continuum error.

Appendix B. Repeated Euler transfer comparisons

Table B.1 retains the higher-order alternatives to the representative main-text comparison. Table B.2 gives all three realizations, including NN1 with global correction. Transfer-integral defects and profile changes are separate measures: cancellation between signed integral changes does not bound local distortion.

Table B.1: Frequency comparison at safety factor $s = 4$ for all transfer variants. Δn is the number of physical steps between mesh changes; $\Delta x/h$ and $\Delta w/h$ are signed position and width changes relative to the fixed-mesh reference.

Transfer	Δn	$\Delta x/h$	$\Delta w/h$	$\sum \Delta I_\rho$
NN1	10	0.14	0.57	$1.1 \cdot 10^{-4}$
NN1	2	0.1	21.41	$5 \cdot 10^{-4}$
NN2	10	−0.02	−0.01	$-8.8 \cdot 10^{-6}$
NN2	2	−0.04	0	$-1.7 \cdot 10^{-5}$
NN3	10	0.01	−0.01	$2.2 \cdot 10^{-5}$
NN3	2	0.02	0	$2.7 \cdot 10^{-5}$
BAR2	10	0.01	0.25	$2 \cdot 10^{-5}$
BAR2	2	−0.03	3.24	$1.4 \cdot 10^{-4}$
BAR3	10	0	0.01	$-1.7 \cdot 10^{-5}$
BAR3	2	0	0.07	$-5.2 \cdot 10^{-6}$
GP2	10	0.01	0	$2.6 \cdot 10^{-6}$
GP2	2	0.01	0	$-3 \cdot 10^{-6}$
GP3	10	−0.01	0	$-2.1 \cdot 10^{-5}$
GP3	2	−0.01	0.02	$1.8 \cdot 10^{-5}$

Table B.2: High-frequency Euler comparison across three random deformation realizations. GC indicates whether the global correction is applied; $s = 2$ permits twice the displacement amplitude of $s = 4$. Position and width changes are in cells of the $h = 0.01$ reference mesh. The final column is the signed sum of target-minus-source density-integral changes at transfer.

Transfer	GC	s	Realization	$\Delta x/h$	$\Delta w/h$	$\sum \Delta I_\rho$
BAR2	No	2	1	-0.06	22.91	$2.7 \cdot 10^{-4}$
BAR2	No	2	2	-0.03	22.98	$3.1 \cdot 10^{-4}$
BAR2	No	2	3	-0.07	22.9	$1.9 \cdot 10^{-4}$
BAR2	No	4	1	-0.03	3.24	$1.4 \cdot 10^{-4}$
BAR2	No	4	2	0	3.2	$1.7 \cdot 10^{-4}$
BAR2	No	4	3	-0.04	3.12	$6.6 \cdot 10^{-5}$
GP2	No	2	1	0.17	0.06	$-1.1 \cdot 10^{-5}$
GP2	No	2	2	0.19	0.06	$-1 \cdot 10^{-5}$
GP2	No	2	3	0.19	0.03	$-9.7 \cdot 10^{-6}$
GP2	No	4	1	0.01	0	$-3 \cdot 10^{-6}$
GP2	No	4	2	0.02	0	$-3.1 \cdot 10^{-6}$
GP2	No	4	3	0.01	0.01	$-2.6 \cdot 10^{-6}$
NN1	Global	2	1	-0.91	25.65	$4.3 \cdot 10^{-14}$
NN1	Global	2	2	-0.66	23.8	$2.7 \cdot 10^{-14}$
NN1	Global	2	3	0	26.32	$1.2 \cdot 10^{-14}$
NN1	Global	4	1	-0.15	18.36	$5.8 \cdot 10^{-14}$
NN1	Global	4	2	-0.01	21.28	$-9.3 \cdot 10^{-16}$
NN1	Global	4	3	-0.12	21.18	$9.4 \cdot 10^{-14}$
NN1	No	2	1	-0.38	26.36	$1.3 \cdot 10^{-3}$
NN1	No	2	2	-0.29	24.98	$-1.6 \cdot 10^{-3}$
NN1	No	2	3	0.44	27.14	$4.9 \cdot 10^{-4}$
NN1	No	4	1	0.1	21.41	$5 \cdot 10^{-4}$
NN1	No	4	2	0.3	22.89	$-8.9 \cdot 10^{-4}$
NN1	No	4	3	0.2	22.75	$8.3 \cdot 10^{-4}$

Appendix C. Controlled-history diagnostics

Appendix C.1. Directional response and boundary enforcement

Table C.1 gives the complete response ratios. The state ratios use the volume-weighted norm; the older-only force ratio is signed. Their agreement with the temporal coefficients is approximate and does not remove the superposition errors.

Table C.1: Euler BDF2 response ratios and relative superposition defects. Subscripts f, n, and $n-1$ denote both-histories, latest-only, and older-only perturbations. Ideal directional ratios are $4/3$ and $-1/3$; the state columns show their nonnegative norm ratios.

Transfer	e^U_{sup}	$\|\delta_n\|_M/\|\delta_f\|_M$	$\|\delta_{n-1}\|_M/\|\delta_f\|_M$	$e^{C_L}_{\mathrm{sup}}$	$\Delta C_{L,n}/\Delta C_{L,f}$	$\Delta C_{L,n-1}/\Delta C_{L,f}$
NN1	$1.8 \cdot 10^{-2}$	1.3333	0.3335	$1.5 \cdot 10^{-2}$	1.3329	-0.3179
NN2	$3.5 \cdot 10^{-2}$	1.3331	0.3341	$4.2 \cdot 10^{-3}$	1.3318	-0.336
NN3	$3.4 \cdot 10^{-2}$	1.3331	0.334	$5.3 \cdot 10^{-3}$	1.3309	-0.3362
BAR2	$3.4 \cdot 10^{-2}$	1.3332	0.334	$3.1 \cdot 10^{-3}$	1.3328	-0.3358
BAR3	$3.3 \cdot 10^{-2}$	1.3332	0.334	$2.8 \cdot 10^{-3}$	1.3329	-0.3357
GP2	$3.4 \cdot 10^{-2}$	1.333	0.334	$3 \cdot 10^{-3}$	1.3322	-0.3352
GP3	$3.4 \cdot 10^{-2}$	1.3331	0.334	$3.3 \cdot 10^{-3}$	1.3326	-0.3359

In the common-trial residual test, the normal momentum mismatch is 2.125×10^{-6} for GP2 and 6.066×10^{-7} for GP3, equal to the normal history forcing at the slip wall. The tangential mismatch is of order 10^{-11}. NN and BAR preserve wall-normal momentum much more closely in this particular transfer. The comparison must apply the same boundary operation to both the measured residual and the predicted history forcing; the raw unconstrained projection is a different state.

Appendix C.2. Cancellation of the initial history forcing

For an imposed target-history discrepancy e, take $\varepsilon^n = e/4$ and $\varepsilon^{n-1} = e$. The initial combination η^n is zero. At a common trial state the two BDF2 equations are then identical, and an exact solve on the same local solution branch should give the same next state. Table C.2 separates the measured force and state diagnostics. First-step lift differences are of order 10^{-9}, but relative state differences are approximately 2×10^{-4} in the Euclidean norm and 0.043–0.090 in the volume-weighted norm. The small force response therefore does not establish cancellation of the complete numerical state, and no independently measured iterative floor is assigned to either norm.

Table C.2: Initial history-cancellation test. Both histories are non-native but $2\varepsilon^n - \varepsilon^{n-1}/2 = 0$. First-step force differences and state differences use the same native continuation; both state norms are shown to avoid interpreting a small load difference as complete state agreement.

Transfer	ΔC_L	ΔC_D	$\|\delta\|_2/\|e\|_2$	$\|\delta\|_M/\|e\|_M$
GP2	$7.67 \cdot 10^{-9}$	$-5.38 \cdot 10^{-9}$	$2.43 \cdot 10^{-4}$	$8.98 \cdot 10^{-2}$
NN1	$2.11 \cdot 10^{-9}$	$-4.1 \cdot 10^{-9}$	$2.1 \cdot 10^{-4}$	$4.34 \cdot 10^{-2}$

The initial residual probes in this additional test use different trial states. For a trial displacement q, their difference has the form

$$\Delta\Phi = \frac{3M_B q}{2\Delta t} + F_B(Y+q) - F_B(Y) - \frac{M_B \eta^n}{\Delta t}. \tag{C.1}$$

With $q = e/4$ and $\eta^n = 0$, the temporal part is three quarters of the older-only forcing magnitude. The measured ratio near 0.75 is therefore not a common-trial verification of Eq. (16): the spatial residual difference also remains. Later cancellation is not expected even in the ideal first-step case. After $\delta^{n+1} = 0$, the older level still contains $e/4$, giving a subsequent history combination $-e/8$.

Appendix C.3. Perturbation-amplitude dependence

Scaling the discrepancy by $\lambda = 1, 1/2, 1/4$ reduces the Euclidean-state and lift superposition remainders for BAR2 and GP2 (Table C.3). The Euclidean absolute state remainder decreases by factors of about 3.2–3.5 per halving, consistent with a higher-than-linear reduction over this finite range. The volume-weighted absolute remainder decreases much more slowly, from approximately 2.4×10^{-5} to 1.5–1.6×10^{-5}, while its normalized value rises from about 3.4% to 8.7–9.2%. These results support directional response in selected diagnostics, but not a uniform quadratic remainder or a verified numerical floor in all norms.

Table C.3: Amplitude dependence of first-step superposition. $\mathcal{E} = \delta_f - \delta_n - \delta_{n-1}$ is the state remainder and $\mathcal{E}_L$ its lift counterpart. Relative remainders use the corresponding both-histories response. The Euclidean and volume-weighted state metrics have different amplitude dependence.

Transfer	λ	$\|\mathcal{E}\|_2$	$\|\mathcal{E}\|_2/\|\delta_f\|_2$	$\|\mathcal{E}\|_M$	$\|\mathcal{E}\|_M/\|\delta_f\|_M$	$\|\mathcal{E}_L\|$
BAR2	1	$1.76 \cdot 10^{-3}$	$7.78 \cdot 10^{-3}$	$2.36 \cdot 10^{-5}$	$3.36 \cdot 10^{-2}$	$2.22 \cdot 10^{-5}$
BAR2	0.5	$5.45 \cdot 10^{-4}$	$4.82 \cdot 10^{-3}$	$1.97 \cdot 10^{-5}$	$5.6 \cdot 10^{-2}$	$4.74 \cdot 10^{-6}$
BAR2	0.25	$1.6 \cdot 10^{-4}$	$2.83 \cdot 10^{-3}$	$1.54 \cdot 10^{-5}$	$8.74 \cdot 10^{-2}$	$1.76 \cdot 10^{-6}$
GP2	1	$1.88 \cdot 10^{-3}$	$8.31 \cdot 10^{-3}$	$2.42 \cdot 10^{-5}$	$3.39 \cdot 10^{-2}$	$2.26 \cdot 10^{-5}$
GP2	0.5	$5.71 \cdot 10^{-4}$	$5.06 \cdot 10^{-3}$	$2.03 \cdot 10^{-5}$	$5.69 \cdot 10^{-2}$	$3.64 \cdot 10^{-6}$
GP2	0.25	$1.64 \cdot 10^{-4}$	$2.91 \cdot 10^{-3}$	$1.65 \cdot 10^{-5}$	$9.22 \cdot 10^{-2}$	$1.53 \cdot 10^{-6}$

Appendix D. RANS numerical sensitivity

Appendix D.1. Inner iterations and time-step refinement

Table D.1 distinguishes achieved residual criteria from iteration limits. The nominal-setting rows cover the full repeated-remeshing histories; the other rows cover the short sensitivity calculations. Raising the SA iteration limit to 1600 reaches the density criterion throughout the sensitivity window. SA also reaches that criterion at every half- and quarter-step point. SST reaches the iteration limit on 28–30 of 38 steps even with a limit of 6400 inner iterations, and on 57–58 of 78 half-step points. Increasing the iteration allowance is therefore not equivalent to obtaining a converged SST reference.

Table D.1: Inner-iteration behavior for the paired calculations of the first deformation realization. Entries separated by a slash are GP2/NN2. $N_{\max}$ is the iteration limit; residuals are median terminal $\log_{10}\mathrm{rms}(r_\rho)$. Nominal-setting rows cover the full interval; the remaining rows cover the short numerical-sensitivity window.

Closure	$N_{\max}$	$\Delta t/\Delta t_0$	Physical steps	Steps at limit	Residual
SA	800	1	398	144/156	-12.00/-12.00
SA	1600	1	38	0/0	-12.00/-12.00
SA	800	1/2	78	0/0	-12.00/-12.00
SA	800	1/4	158	0/0	-12.00/-12.00
SST	800	1	398	333/323	-11.10/-11.24
SST	1600	1	38	28/28	-11.33/-11.33
SST	6400	1	38	28/30	-11.38/-11.28
SST	800	1/2	78	57/58	-11.24/-11.29

Table D.2 reports the sensitivity to increased iteration limits using Eq. (27). SA lift and total-drag responses change by at most 0.3% of their own RMS; their smaller paired differences change by 8.4–10.9%. For SST, the 6400-cap comparison changes the paired lift difference by 68.1% of its nominal-setting RMS. Persistent residual stagnation accompanies this sensitivity, but does not identify a unique cause in the turbulence equations.

For the SA step-refinement sequence, the paired lift RMS is approximately 1.21×10^{-5}, 1.60×10^{-5}, and 2.08×10^{-5}. The corresponding drag values are 2.19×10^{-6}, 3.77×10^{-6}, and 4.55×10^{-6}. The detailed paired response is not settled. Individual RMS amplitudes remain closer across the same samples, but that observation cannot replace a convergence test of the histories. Friction drag is more sensitive than total drag: the SA quarter-step

Table D.2: Effect of increasing the inner-iteration limit, relative to the nominal setting. Columns have the same definition as Table 5: individual values are the larger NN2 or GP2 sensitivity, and paired values refer to their difference. All entries are percentages.

		Individual histories		Paired difference	
Closure	Iteration limit	C_L	C_D	C_L	C_D
SA	1600	0.3	0.2	10.9	8.4
SST	1600	3.7	1.1	13.7	6.0
SST	6400	8.6	2.5	68.1	25.7

history differences are approximately 76–91% of the nominal-setting friction-response RMS. No componentwise convergence claim follows from the smaller total-drag sensitivity.

Appendix D.2. Steady mesh sensitivity and cross-mesh robustness

The coarse, medium, and fine steady meshes in Table D.3 preserve the first off-wall spacing while changing the remaining resolution. Lift, total drag, and pressure drag decrease monotonically, but friction drag does not. The count-based spacing is only a mesh-density descriptor for this anisotropic family. The results do not demonstrate a common asymptotic error expansion, and no extrapolated loads or discretization-uncertainty estimate are reported. Differences in absolute loads remain distinct from errors in a paired transfer comparison on the same mesh.

Table D.3: Steady aerodynamic loads without remeshing on the coarse, medium, and fine viscous meshes. The count-based descriptor $h/h_c = \sqrt{N_c/N}$ uses the coarse-mesh vertex count N_c as reference; it is not a uniform refinement ratio in every direction. The same first off-wall spacing is used on all levels.

Closure	Mesh	Vertices	h/h_c	C_L	C_D	$C_{D,p}$	$C_{D,v}$
SA	Coarse	23,271	1	0.67626	0.010229	0.003559	0.006671
SA	Medium	32,102	0.851	0.67084	0.010048	0.003401	0.006647
SA	Fine	48,228	0.695	0.66854	0.009996	0.00333	0.006665
SST	Coarse	23,271	1	0.67561	0.009858	0.003469	0.006389
SST	Medium	32,102	0.851	0.6686	0.009705	0.003321	0.006385
SST	Fine	48,228	0.695	0.66639	0.009644	0.003247	0.006397

On the medium mesh, the short SA remeshing interval gives individual lift RMS responses near 1.07×10^{-4} and drag responses near 5.39×10^{-5}; the paired differences are 8.77×10^{-6} and 9.94×10^{-7}, respectively. Thus the two methods also remain relatively close on another mesh. Because new node positions produce a different random displacement field, this comparison combines mesh resolution and disturbance realization. The smaller response cannot be attributed to refinement alone. Five to seven of its 38 physical steps also reach the nominal inner-iteration limit.

Data availability

The numerical tables supporting the figures and comparisons, together with the post-processing routines needed to reproduce the reported reductions, are available in the public data repository. The archive does not include every flow-field output from the simulations.

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